

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination : Semester Midterm
 Duration: **1 Hour 20 Minutes**

Semester: Spring 2026
 Full Marks: **50**

CSE421 / EEE465 : Computer Networks

Answer **ALL** questions. (**Pages: 2**)

Figures in the right margin indicate marks.

Name:	ID:	Section:
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Q1 [CO1]	A user is browsing the web while sending an email; how does the transport layer ensure that the incoming segments are delivered to the correct application process on the host? Why can this not be done using the Logical Address?	4
Q2 [CO2]	A student opens a website in the browser. The page loads normally, but the connection is not encrypted. I. Which protocol and method are used for this webpage transfer? II. What is the expected HTTP status code if the page loads successfully? III. At which layer of the Internet protocol stack does the encryption operate if needed?	1 + 1 + 2
Q3 [CO3] [CO2]	Saba's university uses a proxy server with a 55% cache hit rate. Assume: • If the object is in cache, response time = 0.01 sec • If not in cache, response time = 2 sec I. Calculate the average response time of a request. II. Sometimes a proxy server needs to contact the original server using Conditional GET. Explain why using a proxy server is still beneficial.	3 + 3
Q4 [CO2]	A company sets up a mail system for the domain cybercraft.com. All emails should go to mail.server.com. Users should be able to access their email from multiple devices. I. Identify the DNS record types needed. II. If mail.cybercraft.com → 203.0.113.10, write the DNS records clearly. III. Which email retrieval protocol will support multi-device access?	2 + 2 + 2
Q5 [CO2]	A network administrator uses two methods to control congestion: • Method 1: The system checks available resources before allowing new data to enter the network. • Method 2: When congestion occurs, routers send feedback signals to the sender to reduce the transmission rate. I. Identify which method is open-loop and which is closed-loop congestion control. II. Briefly explain how each method works.	3 + 3
Q6 [CO3]	A TCP connection is in the congestion avoidance phase with a congestion window (cwnd) of 16 MSS . The slow-start threshold (ssthresh) is also 16 MSS . The sender receives three duplicate ACKs .	4 + 2

	I. What are the new values of ssthresh and cwnd after this event? II. How will cwnd change after this?	
Q7 Sunbeam loads two HTML pages: Page X over a persistent HTTP connection Page Y over a non-persistent HTTP connection Page X has 18 objects and Page Y has n objects, both excluding the base HTML file. The total RTT count required to load the two pages is the same. The device takes 25 ms to send a TCP request. The HTTP request and response time for an object is 50 ms. [CO3] I. Calculate the value of n [CO3] II. Calculate the total RTT time required to load Page Y. [CO2] III. As the number of objects increases, explain how the performance of persistent HTTP and persistent HTTP with pipelining changes in terms of RTT.		3 + 3 + 3
Q8 A server sends data segments using a Go-Back-N protocol. The 4th data segment has a sequence number of 10500 and an acknowledgement number of 8700. The 2nd data segment is lost and retransmitted after the 5th data segment. Each data segment (of size 100 bytes) sent by the server is requested by the client using an HTTP Request (of size 100 bytes), and the client initially has a buffer size of 1000 bytes. [CO3] I. Calculate the initial sequence numbers (ISN) of both the client and the server. II. What acknowledgement number will the receiver send before receiving the retransmitted 2nd segment? III. After receiving the retransmitted 2nd segment, determine the RWND of the receiver.		4 + 3 + 2